MIT 18.06SC - 2.0

Problem 9.1

v_1, v_2 and v_3 are independent (~~Every~~ each vector has a non zero entry for which the others have zero entries hence not a linear combination of the others).
$v_4 = v_2 - v_1$, $v_5 = v_3 - v_1$ and $v_6 = v_3 - v_2$, so all the remaining vectors are dependent on v_1, v_2, v_3 hence the maximum number of independent vectors among the group is 3.

Problem 9.2

The equation can be expressed in the form $Ax = b$ as:-

$$\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Setting each free variable to 1 and the other to 0 we get:-

$$\begin{bmatrix} x \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} x \\ 0 \\ 1 \end{bmatrix}$$

Solving both for x:-

$x - 2(1) + 3(0) = 0$ $\quad$ $x - 2(0) + 3(1) = 0$
$x = 2$ $\quad$ $x = -3$

So the basis for the given plane are the vectors:-

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Because the given plane is spanned by these vectors.

The xy plane contains all vectors for which $z=0$, and so it contains $[2 \ 1 \ 0]^T$. Since the intersection of the given plane and the xy-plane is a line, every vector on it is hence a multiple of $[2 \ 1 \ 0]^T$ and so $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ is a basis for it.

~~He~~ We need a vector v such that $v_1 \cdot v = 0$ and $v_2 \cdot v = 0$ i.e $2x+y=0$ and $-3x+z=0$ (where x, y, z are the entries of v). ~~So~~ Let $x=1$, the $y=-2$ and $z=3$, so $[1 \ -2 \ 3]^T$ is the basis for all vectors perpendicular to the given plane as its multiples 'cv' describe that entire space.